

Tropical Cyclones

Large-Scale Conditions for Formation

Purpose. As we work through the lecture, use this sheet to distinguish three things: (1) what we observe about where and when tropical cyclones form, (2) environmental conditions that are associated with formation, and (3) possible physical reasons those conditions matter. A recurring question is whether a condition is *necessary*, *sufficient*, or simply *statistically favorable*.

Learning objectives. By the end of class, you should be able to:

1. describe the major large-scale environmental conditions historically associated with tropical cyclogenesis;
2. explain why warm SST alone does not determine whether a tropical cyclone forms;
3. calculate deep-layer vertical wind shear and explain two ways shear can inhibit genesis;
4. explain why mid-tropospheric moisture and a pre-existing disturbance can favor genesis;
5. interpret the structure of a genesis potential index and explain why such indices are useful.

Key reference from the lecture: Emanuel, K., 2018: *100 Years of Progress in Tropical Cyclone Research*.

1. Where and When Do Tropical Cyclones Form?

Slides 3–4

Look at the global tropical-cyclone track/genesis map shown in class. Record observations *before* trying to explain them.

A. Geographic pattern

Typical latitude range of formation: _____

Basins with frequent formation: _____

Regions with relatively little formation: _____

One pattern that surprised you: _____

In-class exercise. From the map alone, propose one environmental variable that might help explain the geographic distribution of tropical cyclogenesis. State your hypothesis in one sentence.

B. Three big questions from the lecture

1. Why do tropical cyclones form in particular regions of the world?
2. Why does each basin have a preferred hurricane/typhoon season?
3. Roughly 100 tropical cyclones form globally each year. What controls the global number?

Choose one of these questions and write a preliminary explanation based on what you already know.

2. The Historical “Favorable Environment”**Slides 5–6**

The lecture begins with the influential framework associated with Gray (1968) and later studies. The commonly cited large-scale conditions are:

- sea-surface temperature (SST) exceeding roughly 26°C;
- divergent flow in the upper troposphere;
- a moist lower-to-middle troposphere;
- a pre-existing cyclonic circulation or disturbance;
- low vertical wind shear.

Do not treat this as a checklist that guarantees genesis. Use the table to record the physical reason discussed for each condition.

Environmental factor	Why might this favor tropical cyclogenesis?
Warm SST	
Upper-level divergence	
Moist lower/middle troposphere	
Pre-existing cyclonic disturbance	
Low vertical wind shear	

In-class exercise. For each factor above, label it in the margin as one of the following: *clearly necessary*, *often favorable*, or *uncertain / context dependent*. Be prepared to revise your labels as the lecture proceeds.

Class notes / revisions

3. Sea-Surface Temperature

Slides 7–8

In the lecture, we cited two papers: Tory and Dare, 2015 and McTaggart-Cowan et al. 2015. Based on these papers, we emphasized four observations about SST:

- A climatological SST near 26°C approximately outlines many regions where tropical cyclones frequently form.
- There are basin-to-basin differences, and some storms form over colder water.
- The frequency of genesis increases smoothly with increasing SST; there is no abrupt jump at exactly 26°C .
- Even over water warmer than 26°C , tropical cyclone formation remains rare.

Key distinction. A useful empirical threshold is not necessarily a sharp physical switch. Warm SST may help create a favorable thermodynamic environment, but warm water by itself does not guarantee genesis.

A. In your own words, explain why the statement “Tropical cyclones require $\text{SST} > 26^{\circ}\text{C}$ ” is an oversimplification.

In-class exercise. Suppose two large ocean regions both have $\text{SST} = 29^{\circ}\text{C}$. Tropical cyclones form frequently in Region A but rarely in Region B. List at least three other environmental differences you would investigate.

B. Threshold versus probability

Complete the statement:

Increasing SST tends to make genesis more _____, but does not make genesis _____.

Notes from discussion

4. Vertical Wind Shear

Slides 9–11

Deep-layer vertical wind shear is commonly defined as the magnitude of the vector wind difference between an upper and a lower level:

$$S = [(U_2 - U_1)^2 + (V_2 - V_1)^2]^{1/2}$$

In the lecture, level 2 is typically 200 hPa and level 1 is typically 850 hPa.

A. What does the formula mean physically?

Wind shear depends on the *vector difference*, not simply on the difference between wind speeds. Both wind direction and wind magnitude can matter.

In-class exercise. At 850 hPa, $(u, v) = (5, 0)$ m s⁻¹. At 200 hPa, $(u, v) = (5, 10)$ m s⁻¹. Calculate the 850–200-hPa shear magnitude.

B. Empirical behavior discussed in the lecture

- Large vertical shear is generally unfavorable for tropical cyclogenesis.
- The lecture cites work suggesting genesis is unlikely when 850–200-hPa shear exceeds roughly 15 m s⁻¹.
- Results discussed from Nolan and McGauley indicate that light shear (about 2.5–5 m s⁻¹) can be especially favorable; near-zero shear is not necessarily unfavorable but is uncommon.
- Their simulations also suggest sensitivity to shear direction.

Question: Why is “zero shear is always best” not the same statement as “large shear is unfavorable”?

C. Physical mechanisms

The lecture highlights two broad ways that shear can interfere with a developing cyclone.

1. **Ventilation effect:** shear can introduce unsaturated environmental air into the developing core. Evaporation of precipitation can promote cool downdrafts and bring lower-entropy/drier air toward the boundary layer.
2. **Dynamical effect:** shear can displace upper- and lower-level circulations, producing a vertically tilted structure.

For each mechanism, write the consequence for organized deep convection.

Ventilation: _____

Vortex tilt: _____

In-class exercise. Sketch a vertically aligned developing vortex and a strongly sheared/tilted vortex. Add arrows showing the environmental shear.

Notes / sketch:

5. Mid-Tropospheric Moisture

Slide 12

The lecture emphasizes sensitivity of tropical cyclogenesis to moisture in the middle troposphere, around 600 hPa.

Sub-saturated midlevels can promote entrainment of dry air within developing thunderstorms. This leads to evaporation of precipitation and cold downdrafts which makes the boundary layer less favorable for sustained convection.

A. Explain the role of evaporation. Why can precipitation falling through dry air make subsequent convection more difficult?

B. Connect moisture and wind shear. How can strong vertical shear make a dry mid-troposphere more damaging to a developing cyclone?

In-class exercise. Consider two disturbances with the same SST, vorticity, and vertical shear. One has a moist mid-troposphere and the other has a dry mid-troposphere. Predict which is more likely to sustain repeated deep convection and explain why.

6. Pre-Existing Disturbances

Slides 13–16

Although, in theory, tropical cyclones can emerge spontaneously from undisturbed atmosphere (more on this in a subsequent section on TC genesis), in the present climate, tropical cyclones invariably form within a pre-existing cyclonic disturbance. Precursors listed in the lecture include:

- African easterly waves in the Atlantic;
- the ITCZ in the eastern Pacific;
- equatorial waves in the western Pacific;
- monsoon troughs/depressions in the Indian Ocean and Bay of Bengal.

A. Why might a pre-existing cyclonic disturbance help?

Write down two possible advantages of starting with an already organized disturbance.

1. _____
2. _____

B. African easterly waves

As the AEW/genesis figures are shown, record what you infer about the relationship between easterly waves and Atlantic tropical cyclogenesis.

Notes / sketch:

In-class exercise. Does an association between AEWs and tropical cyclogenesis prove that AEWs are dynamically required for genesis? Explain the distinction between *common pathway* and *strict requirement*.

C. Is a precursor absolutely required?

The lecture notes that idealized research has demonstrated tropical-cyclone-like vortices emerging through self-aggregation of convection from an initially unorganized environment.

What does this result change about the statement “a pre-existing disturbance is necessary for tropical cyclogenesis”?

7. Putting Environmental Controls Together

Slides 17–20

At this point we have several environmental variables:

SST / thermodynamics, moisture, vorticity, vertical shear.

A natural question is whether their combined influence can be summarized in one diagnostic quantity.

A. Why combine variables?

List two potential advantages of constructing a combined genesis index.

1. _____
2. _____

In the lecture, we reviewed empirical genesis indices beginning with Gray (1975, 1979), followed by later formulations such as Emanuel and Nolan (2004), Emanuel (2010), Tippett et al. (2011), Bruyère et al. (2012), and others.

Important. A genesis potential index is an environmental diagnostic. It summarizes how favorable the large-scale environment appears; it is not a deterministic prediction that a storm must form.

8. Emanuel–Nolan Genesis Potential Index

Slide 20

The form shown in the lecture is

$$GP = |10^5 \eta|^{3/2} \left(\frac{RH}{50} \right)^3 \left(\frac{V_{\text{pot}}}{70} \right)^3 (1 + 0.1 V_{\text{shear}})^{-2}$$

where

- η = absolute vorticity at 850 hPa;
- RH = relative humidity at 700 hPa;
- V_{pot} = potential intensity (more on this soon);
- V_{shear} = 850–200-hPa vertical wind shear magnitude.

A. Read the equation qualitatively.

Without doing any arithmetic, indicate what happens to GP when each quantity increases.

Variable increases	Expected change in GP	Favorable or unfavorable?
$ \eta $		
RH		
V_{pot}		
V_{shear}		

In-class exercise. Which variables enter with positive powers? Which enters with a negative power? Explain what this says about the empirical environmental relationships represented by the index.

B. Sensitivity exercise

If RH increases from 50% to 60% while everything else remains fixed, by what multiplicative factor does the humidity term change?

$$\left(\frac{60}{50}\right)^3 = \underline{\hspace{2cm}}$$

If vertical shear increases, does the shear factor increase or decrease? Explain using the exponent -2 .

C. What is not explicit in this equation?

List one process or structural property of a developing disturbance that is not directly represented by these four large-scale environmental variables.

9. Comparing Genesis Potential with Observed Storm Counts

Slides 21–22

In the lecture, we examined basin-averaged genesis potential together with annual tropical cyclone counts, and scatterplots relating the two.

As you view the figures, record the following.

Does higher GPI generally correspond to more storms? _____

Is the relationship perfectly deterministic? _____

Which basin appears to show the clearer relationship? _____

What sources of scatter might exist? _____

In-class exercise. Suppose two years have nearly identical basin-mean GPI but very different storm counts. Give two reasons why this does not automatically invalidate the GPI concept.

10. Why Are Genesis Potential Indices Useful?

Slides 23–24

Applications listed in the lecture include:

- seasonal forecasting of tropical cyclone activity;
- understanding the combined effect of environmental controls on genesis;
- benchmarking numerical-model environments against observations;
- diagnosing possible changes in tropical cyclone frequency in future-climate simulations.

For each application, write one sentence describing what the index contributes.

Seasonal forecasting: _____

Process understanding: _____

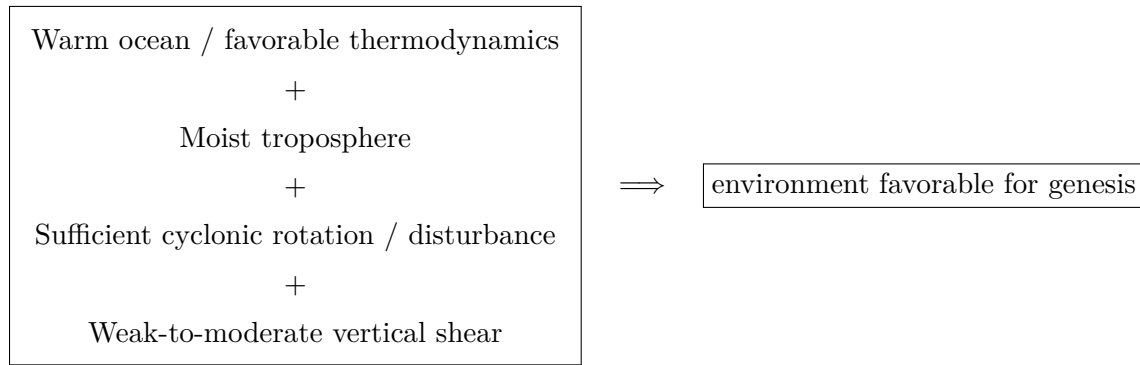
Model evaluation: _____

Climate-change studies: _____

In-class exercise. What is one danger of treating a genesis potential index as though it were a complete physical theory of tropical cyclogenesis?

11. Synthesis: Favorable Does Not Mean Guaranteed

Complete the conceptual chain below.



But

favorable environment $\neq$ guaranteed tropical cyclone

A. Why not?

B. Necessary versus sufficient

In your own words:

Necessary condition: _____

Sufficient condition: _____

Favorable condition: _____

In-class exercise. Return to the five environmental factors in Section 2. Would you change any of your original labels after the lecture? Identify one change and explain it.

12. Exit Questions

Answer briefly before next class.

1. Why is 26°C better viewed as an empirical guide than as a sharp switch for genesis?

2. Give one thermodynamic and one dynamical mechanism by which vertical shear can inhibit genesis.

3. Why does mid-tropospheric dry air make persistent deep convection more difficult?

4. What role can a pre-existing disturbance play, and why is it not necessarily an absolute requirement?

5. In one sentence, what does a genesis potential index actually tell us?

References highlighted in the lecture: Gray (1968, 1975, 1979); Bister and Emanuel (1997); DeMaria et al. (2001); Davis and Bosart (2003); Emanuel and Nolan (2004); Emanuel (2010); Nolan (2011); Tippett et al. (2011); Korty et al. (2012); Bruyère et al. (2012); Tory and Dare (2015); McTaggart-Cowan et al. (2015); Emanuel (2018).